

Chapter 3

Vectors

A vector is a quantity that has direction as well as magnitude. Quantities that only have magnitude are called scalars.

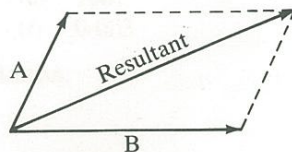
Vectors	Scalars
displacement	distance
velocity	speed
acceleration	time
momentum	mass
force	temperature
moment	charge
flux density	density
electric field intensity	volume
	area
	energy
	power

All vector quantities must involve a statement of direction. Vectors can be represented by an "arrow" in which the magnitude of the vector is given by its length and the direction is given by the direction of the arrow.

Addition

The addition of vectors must take into account the directional factor and the result is best seen in a diagram.

All vectors are added by placing the tail of one to the tip of the first. The resultant is the vector which goes from the tail of the first vector added to the tip of the last added. The parallelogram rule is often used for adding vectors that act simultaneously. If force vector A is to be added to force vector B the resultant is as shown.

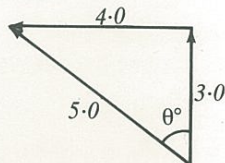


When vectors act at right angles to each other, the diagonal is easily found by Pythagoras' Theorem. The direction of the resultant is also easily found by trigonometry.

Example 1

What is the sum of two displacements 3.0 m north and 4.0 m west?

$$\theta = \tan^{-1} \left(\frac{4}{3} \right) = 53^\circ$$



The resultant displacement is 5.0 m at north 53° west.

Subtraction

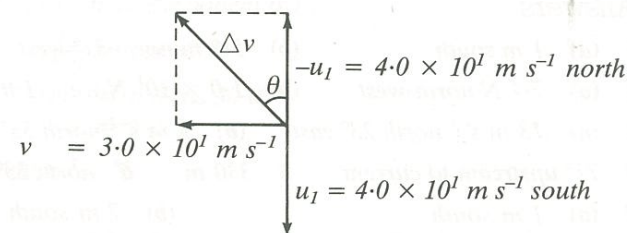
value of a vector is another vector with the same magnitude but having the opposite direction.

$$A - B = A + (-B)$$

Example 2

What is the change in velocity when a cricket ball, moving at $4.0 \times 10^1 \text{ m s}^{-1}$ south, is hit over square leg which is due west at $3.0 \times 10^1 \text{ m s}^{-1}$?

$$\begin{aligned} u &= 4.0 \times 10^1 \text{ m s}^{-1} \text{ south} & \Delta v &= v - u = v + (-u) \\ v &= 3.0 \times 10^1 \text{ m s}^{-1} \text{ west} & \Delta v &= 3.0 \times 10^1 \text{ m s}^{-1} \text{ west} + 4.0 \times 10^1 \text{ m s}^{-1} \text{ north} \\ \Delta v &= ? \end{aligned}$$



$$\begin{aligned} \Delta v &= \sqrt{30^2 + 40^2} = 50 \\ \theta &= \tan^{-1} \left(\frac{30}{40} \right) = 37^\circ \end{aligned}$$

The change in velocity is $5.0 \times 10^1 \text{ m s}^{-1}$ at north 37° west.

Set 3

- Add the following displacements.
 - 3 m north and 4 m south
 - 5.0 m north and 3.0 m west
- Add the forces:
 - 5.0 N west 5.0 N north
 - 6.0 N west + 6.0 N south + 6.0 N north-west
- Find the sum of the following velocities.
 - 12 m s⁻¹ north and 5.0 m s⁻¹ east
 - 6.0 m s⁻¹ north, 4.0 m s⁻¹ east, 3.0 m s⁻¹ south
- A boat can travel at 3.0 m s⁻¹ in still water and wishes to travel directly across a river with a current of 1.0 m s⁻¹. What direction upstream does the boat need to be steered?
- If the boat in question 4 heads directly across the river which is 1.0 km wide how far downstream will it reach the opposite bank?
- A plane is heading due north at 120 km h⁻¹ in an easterly crosswind of 50.0 km h⁻¹. What is the actual direction along the ground?
- Subtract the following vectors.

- 8 If a car travelling south at 40 m s^{-1} slows down to 20 m s^{-1} , what is the change in velocity?
- 9 A batsman hits a cricket ball travelling towards him at 30 m s^{-1} over square leg at 50 m s^{-1} . If the cricketer is facing north and square leg is west of him, what is the change in velocity of the ball?
- 10 If a car travelling south at 50 km h^{-1} rounds a bend and travels east at the same speed, what is its change in velocity?

Answers

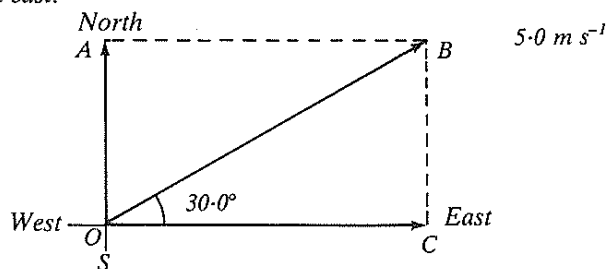
- 1 (a) 1 m south (b) 5.8 m north 31° west
- 2 (a) 7.1 N north-west (b) $1.0 \times 10^1 \text{ N}$ west (1.0×10^1)° south
- 3 (a) 13 m s^{-1} north 23° east (b) 5 m s^{-1} north 53° east
- 4 71° upstream to current 5 330 m 6 north 23° west
- 7 (a) 1 m south (b) 7 m south
(c) 5 m south 40° east (d) 5.6 m south 31° west
- 8 20 m s^{-1} north 9 60 m s^{-1} north 60° west
- 10 20 m s^{-1} north-east

Resolution of vectors

Just as two vectors can be added, a single vector can be split into two parts or components. It is often easier to deal with the components of several vectors in two mutually perpendicular directions, since the components in one direction have no effect on the component in the other direction.

Example 3

What are the components of a velocity of 5.0 m s^{-1} north 60.0° east in the directions north and east?



The two components must add up to the origin vector. Thus completing the rectangle OABC the components are north, OA, and east, OC.

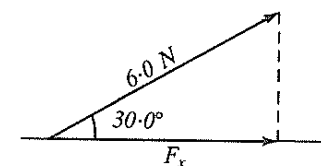
Using trigonometry $OB = 5.0 \text{ m s}^{-1}$ $\angle BOC = 30.0^\circ$

north component $= 5.0 \sin 30.0^\circ = 2.5 \text{ m s}^{-1}$

east component $= 5.0 \cos 30.0^\circ = 4.3 \text{ m s}^{-1}$

Example 4

What is the horizontal component of a force of 6.0 N acting at 30.0° to the horizontal?



horizontal component is F_x

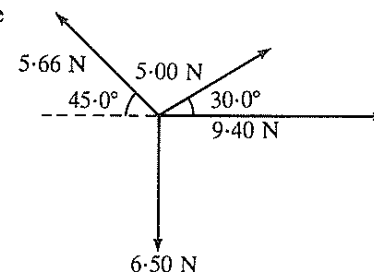
$$\frac{F_x}{6.0} = \cos 30.0^\circ$$

$$\therefore F_x = 6.0 \cos 30.0^\circ = 5.2 \text{ N}$$

The horizontal component is 5.2 N.

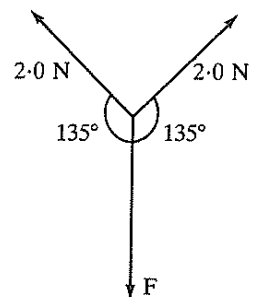
Set 4

- 1 What is the horizontal component of a force of 11 N acting at 60.0° to the horizontal?
- 2 What is the northerly component of a wind blowing at 15 m s^{-1} from the south-east?
- 3 What is the vector whose components are 3 N horizontally and 4 N vertically?
- 4 What are the vertical and horizontal components of a force of 25 N acting at 30.0° to the horizontal?
- 5 A plane taking off leaves the runway at 32° to the horizontal, travelling at 180 km h^{-1} . How long will it take to climb to an altitude of 1000 m?
- 6 What is the acceleration of a small object down a smooth plane inclined at 20.0° to the horizontal? Take the acceleration due to gravity $g = 9.80 \text{ m s}^{-2}$.
- 7 (a) Add the vertical components of the forces acting on the point shown in the diagram.
(b) Which way will it move?
- 8 What is the total force acting in question 7?



- 9 A boat crossing a river which flows at 5.0 m s^{-1} south heads directly across at 2.0 m s^{-1} . If the river is 1.0 km wide:

10



What is F if the sum of all the forces in the diagram is zero?

SECTION 2

MECHANICS

Answers

1 5.5 N

2 11 m s^{-1} 3 5 N 50° above the horizontal

4 12 N vertical, 22 N horizontal

5 42 s

6 3.35 m s^{-2}

7 (a) 0.0 N

(b) to the right

8 9.73 N to the right

9 (a) $5.0 \times 10^2 \text{ s}$ (b) $2.5 \times 10^3 \text{ m}$ 10 2.8 N at 135° to both 2.0 N forces
